Multi-vector attack correlation, SSH brute force, and web reconnaissance

EXECUTIVE SUMMARY

The investigation identified malicious activity against LearnCyber surfaces from 195.178.110.228. This activity included repeated failed SSH authentication attempts against the root account from 195.178.110.228, and Cloudflare recorded and blocked a separate malicious request from 195.178.110.15 targeting /wp-config.php-backup.
The events occurred approximately three seconds apart, and the investigated IPs shared infrastructure associated with AS48090. The evidence supports broader reconnaissance or multi-vector malicious activity, with medium confidence. No successful SSH authentication or successful retrieval of the targeted web resource was identified in the reviewed evidence.

INVESTIGATION WORKFLOW

ALERT TRIAGE

Investigation date: 22-09-2026
Incident date: 04-09-2026 14:30:42
Primary SSH source IP: 195.178.110.228
Environment: LearnCyber
Web attack source IP: 195.198.110.15
Primary Host: SOC
Severity: 5
Target system: soc
Target account: root
Authentication method: SSH
Authentication result: Failed
Rule description: PAM: user login failed

Host Investigation

There wasn't a single failed login event, but rather multiple attempts across several minutes. There was no password spraying, but rather a single root account was hit repeatedly. This confirms brute force, linked to password guesses against one account.
There was partly automated scanning, as the mechanics are clearly **automated/bot-driven** (rapid connect-fail-disconnect, sequential source ports, machine-like timing), and lastly, **no compromise**

Time	Source	IP	Activity

14:25:39	Attacker	195.178.110.228	Connection closed on Port 33318 (initial probe)
14:28:57	Attacker	195.178.110.228	Authentication failure
14:29:00	Attacker	195.178.110.228	Failed password for root on port 36664
14:29:01	Attacker	195.178.110.228	Connection closed by authenticating root user on port 36664
14:30:42	Attacker	195.178.110.228	Authentication failure
14:30:44	Attacker	195.178.110.228	Failed password for root via port 36664
14:30:44	Attacker	195.178.110.228	Connection closed by authenticating user root via port 39892
14:32:32	Attacker	195.178.110.228	Authentication failure
14:32:35	Attacker	195.178.110.228	Failed password for root via port 43078
14:32:36	Attacker	195.178.110.228	Connection closed by authenticating root user via port 43078 [preauth]
14:33:40	Analyst	Localhost	sudo grep 195.178.110.228 /var/log/auth.log- Investigation
14:34:16	Analyst	Localhost	sudo grep 195.178.110.228 /var/log/auth.log- Investigation
Post-Investigation	Analyst	Localhost	sudo ufw deny from 195.178.110.228 to any – Rule added for containment

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Threat investigation

The investigation examined three IP addresses using VirusTotal, GreyNoise, AbuseIPDB, and Shodan.

Investigation analysis for **195.178.110.228** and indicators identified

Source	**Findings**
IP reputation	15/89 security vendors flagged as malicious
ASN	AS48089
ISP	TECHOFF SRV LIMITED
Abuse confidence score	100%
Country shown by AbuseIPDB	Netherlands
AbuseIPDB	52,135 reports from 1775 distinct sources
Scanner categories	Brute force, SSH, hacking, port scan, web app attack
Operating system shown by Shodan	Linux

Investigation analysis for **195.178.110.30** and indicators identified

Source	**Findings**
IP reputation	10/89 security vendors flagged as malicious
ASN	AS48090
ISP	TECHOFF SRV LIMITED
Abuse confidence score	100%
Country shown by AbuseIPDB	Netherlands
AbuseIPDB	422,921 reports from 596 distinct sources
Scanner categories	Brute force, SSH, port scan, web app attack
Operating system shown by Shodan	Ubuntu

Investigation analysis for **195.178.110.15** and indicators identified

Source	**Findings**
IP reputation	14/89 security vendors flagged as malicious
ASN	AS48090
ISP	TECHOFF SRV LIMITED
Abuse confidence score	100%
Country shown by AbuseIPDB	Netherlands
AbuseIPDB	178,665 reports from 1,836 distinct sources
Scanner categories	Brute force, hacking, port scan, bad web bot, web app attack
Operating system shown by Shodan	Linux

MITRE ATT&CK MAPPING

- **Password Guessing – T1110.001**

Status: Confirmed
Evidence: The authentication logs show three clear attempts: repeated SSH password authentication failures against the root account.
This is the strongest MITRE ATT&CK mapping in the investigation.

- **Active scanning—T1595**

Status: Potentially Applicable
Evidence: Threat intelligence associated 195.178.110.15 with scanning-related activity, including port scanning. However, threat intelligence classification alone does not prove that active scanning occurred against LearnCyber. As such, T1595 is considered potentially applicable, but additional LearnCyber telemetry would be required to confirm active scanning against LearnCyber.

- **Exploit public-facing application—T1190**

Status: Potential but not confirmed
Evidence: The Cloudflare event involved a request for /wp-config.php-backup but was blocked by a Cloudflare rule associated with broken access control and file inclusion. However, requesting a potentially sensitive file does not by itself establish successful exploitation. Therefore, T1190 should not be presented as a confirmed technique; it is only potentially applicable to the web attack if the evidence supports an attempted exploitation pathway, rather than treating every malicious URL probe as a successful exploit.

CONTAINMENT RECOMMENDATION
Immediate actions

- Confirm in the logs that no SSH brute force password guessing is ongoing.
- Confirm all authentication status.
- Monitor and review privileged account activity.
- Block confirmed malicious IP addresses.
- Continue monitoring Cloudflare events.

Short actions
Hardening SSH

- Disable root login over SSH: Prevent direct root access to reduce the attack surface.
- Switch to SSH key authentication only: Turn off password-based logins entirely and require SSH keys for all users.
- Set up automatic lockout after repeated failed attempts: Use tools like fail2ban to temporarily block IPs that show suspicious login behaviour.

- Consider placing SSH behind a VPN: If operationally feasible, restrict SSH access to VPN users only, rather than relying solely on subnet blocking.
- Review your UFW firewall rules: Ensure the default policy is to deny incoming traffic, confirm the block rule for 195.178.110.0/24 is active, and verify that SSH access is limited to trusted IPs only.

Web Security

- Remove backup and configuration files from the web root: Delete any **.bak, .old, .config**, or similar files that could expose sensitive information.
- Restrict origin traffic to Cloudflare IP ranges: Configure your server to accept web traffic only from Cloudflare's IPs so attackers can't bypass the WAF by targeting your origin IP directly.
- Keep Cloudflare WAF rules in block mode: Ensure managed rules are actively blocking threats, not just logging them.
- Enable rate limiting on common scanner paths: Add rate limits for paths frequently targeted by automated scanners (e.g., /wp-admin, /xmlrpc.php).
- Rotate sensitive credentials as a precaution: Change your WordPress database password and regenerate WordPress salts/keys to invalidate any potentially compromised secrets.

Monitoring

- Correlate Cloudflare and SSH logs by source IP and ASN: Combine telemetry from both systems to spot coordinated, multi-vector attacks from the same subnet earlier, similar to the activity seen in this incident.

Long-Term Recommendations

The investigation identified an opportunity to improve the correlation between security controls. Recommended changes include:

1. Develop Wazuh rules for repeated authentication failures against privileged accounts.
2. Correlate SSH authentication events with source IP reputation.
3. Forward relevant Cloudflare security telemetry into the SIEM.
4. Create correlation rules for simultaneous web and infrastructure attacks.
5. Enrich security alerts automatically with threat intelligence.
6. Monitor for repeated activity from infrastructure associated with previously malicious sources.
7. Establish a documented incident response procedure for multi-vector attacks.

Final Conclusion

The investigation identified malicious activity targeting multiple LearnCyber attack surfaces. The first activity involved repeated failed SSH authentication attempts from 195.178.110.228 targeting the privileged root account on the SOC system. The authentication logs support a classification of repeated SSH password guessing, with no successful SSH authentication identified in the reviewed evidence. A separate Cloudflare event involving 195.178.110.15 occurred approximately three seconds after the Wazuh SSH event. The source attempted to access /wp-config.php-backup, a resource that could potentially expose sensitive WordPress configuration information if publicly accessible. Cloudflare

blocked the request before it reached the origin. Threat intelligence investigation identified significant malicious reputation associated with the investigated IP addresses and common infrastructure through AS48090 and TECHOFF SRV LIMITED. Taken together, the evidence supports the assessment that the activity represents broader reconnaissance or multi-vector malicious activity against LearnCyber, rather than an isolated SSH event. The assessment is assigned medium confidence because the temporal, behavioral, and infrastructure correlations are meaningful, but the available evidence does not conclusively establish that all activity originated from the same threat actor.

Final Analyst Verdict: Broader reconnaissance/multi-vector malicious activity against LearnCyber, medium confidence. No successful SSH compromise or successful retrieval of the targeted WordPress backup file was identified in the reviewed evidence.